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# Assessment of $CO_2$ geological storage capacity of saline aquifers under the North Sea

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ABSTRACT

In this study, potential storage sites in the subsurface of the North Sea are analyzed to estimate the available $CO_2$ storage capacity. In total 441 sites in five countries (United Kingdom, Denmark, Norway, Netherlands and Germany) are studied in terms of geophysical characteristics and potential capacity. Instead of a simple volume-based approach that tends to over-estimate capacity, we consider storage limited by the time-dependent pressure increase in each injection well. We used literature data to estimate the geological and physical properties of the storage formations. The study confirmed a tremendous potential for $CO_2$ storage approaching 440 Gt with a possible injection rate of 22 Gt $yr^{-1}$. The United Kingdom can potentially store more than 230 Gt of $CO_2$ over 30 years, a value 20 times higher than its current emissions, while the same scales apply for the Netherlands with 147 Gt $CO_2$ potential. The thirteen oil and gas fields examined in Denmark are able to store around 4 Gt of $CO_2$ over three decades at a rate of 127 Mt $yr^{-1}$ covering more than twice the current emissions, while Norway can store 48 Gt from just 10 large saline aquifers. Despite the potential uncertainties in the data, there is sufficient capacity for $CO_2$ storage to play a significant role in Europe to avert a temperature increase of more than 1.5 °C this century.

## 1. Introduction

According to the Intergovernmental Panel for Climate Change (IPCC, 2021), human activities are estimated to have caused approximately 1.09 [0.95 to 1.20] °C of global warming above pre-industrial levels. This amount is expected to reach 1.5 [1.2 to 1.7] °C between 2021 and 2040 (SSP1), with existing growth rates. Current dependence on fossil fuels and their unconstrained extraction are expected to consume the available carbon budget within the next decade further contributing to global warming (Karvounis and Blunt, 2021). This temperature increase is expected to lead to extreme natural phenomena including extensive droughts, hot days and cold nights near the equator and substantial sea level rise (Masson et al. 2018, Masson-Delmotte et al., 2021). According to the Shared Socioeconomic Pathways (Riahi et al., 2017) an increase of two billion in the population is almost inevitable over the next few decades. In the event of such a scenario, demand for energy and other resources is projected to increase with a consequent increase in greenhouse gas emissions up until 2060 at least.

Carbon capture and Storage (CCS) is expected to be one of the game changers in efforts to combat climate change. Even in the most pessimistic scenarios for the growth of emissions (SSP5, SSP2), CCS plays a significant role with more than 5 Gt per year to be deposited underground in the next 50 years. Since 2009, the European Union has set guidelines for the implementation of large-scale CCS initiatives within its territory, by publishing the directive for carbon capture and storage (2009/31/EC, 2009), urging the 27 member nations to initiate studies and evaluate their storage capacity potential. But what is the potential storage capacity?

For the subsurface, Fang et al. (2010) proposed a set of basic characteristics in saline aquifers that need to be examined prior to injection that will assist with a capacity estimation. These include potential faults in the rock (Alonso et al., 2012), cap rock integrity, and other geological assessments to provide information about $CO_2$ migration, the size of the reservoir, the associated depths, as well as several geophysical characteristics such as the porosity and permeability of the rock. Other important properties to be acquired include fluid properties encompassing brine salinity, water and $CO_2$ viscosity and density that affect its solubility. Schulze et al. (2009) and Heitmann et al. (2012) demonstrated that Europe's emissions are concentrated in the North, mainly due to industrial activity. Therefore, North Sea storage sites have been traditionally investigated (Chadwick et al., 2004; Holloway et al., 2006),

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**Nomeclature**

*Greek Symbols*

| | |
|---|---|
| $\mu_c$ | $CO_2$ viscosity |
| $\mu_w$ | Brine viscosity |
| $\rho_{CO2}$ | $CO_2$ density |
| $\phi$ | Porosity |
| $\Psi$ | Radius of a fictious equivalent vertical interface |
| A | Reservoir area |
| $B_0$ | Oil formation volume factor |
| c | Rock compressibility |
| d | Inter-well distance |
| E | Storage efficiency |
| H | Reservoir thickness |
| k | Absolute permeability |
| $M_{CO2}$ | Mass of $CO_2$ |
| n | Number of wells |
| $P_0$ | Pressure at the top of the reservoir |
| $\Delta P$ | Pressure variation |
| $\Delta P_r$ | Pressure response for reference flowrate |
| Q | Volumetric Flowrate |
| $Q_M$ | Maximum sustainable flowrate |
| $Q_M^{tot}$ | Total injection flowrate referring to the sum of all wells |
| $Q_r$ | Reference flowrate |
| r | Distance from injection well |
| R | Radius of pore pressure propagation |
| STOIIP | Stock Tank Oil Initially in Place |
| t | Time |
| $T_0$ | Temperature at the center of the reservoir |

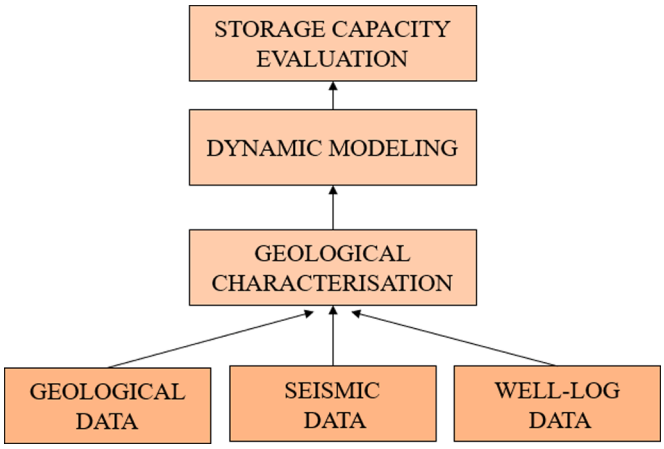



**Fig. 1.** Overview of workflow to determine storage capacity.

in terms of geophysical characteristics (Taylor et al., 1994, 2015). Kolster et al. (2018), used a commercial software to calculate the sensitivity of $CO_2$ storage by varying the injection rate. The Bunter sandstone formation of North Sea was used, while locations of low and medium permeability were chosen as injection sites. Agada et al. (2017) studied the impact of the number of injection sites and injection rate on ultimate $CO_2$ storage. Again, the great Bunter sandstone formation in UK's territory of North Sea was selected. They showed that depth plays a vital role in overall capacity, as it is associated with pressure limits allowed at the storage site. Anthonsen et al. (2014) studied storage in the Baltic Sea, while Heinemann et al. (2012) also focused on Bunter sandstone formations, Bentham (2006) considered the Southern North Sea, while Van der Meer et al. (2009) studied several saline aquifers in the Dutch territory of the North Sea. Other studies, such as Aminu and Manovic, (2020) include the effects of impurities in $CO_2$ streams ($NO_2$, $SO_2$) on reservoir performance. Injectivity is affected only during the early stages of injection into the reservoir.

In this paper, a first order assessment of $CO_2$ geological storage under the North Sea is conducted. While, as outlined above, other studies have been performed during the past decade, these have been country focused or limited to certain geological formations. In contrast, our study examines a total of 441 potential fields categorized according to their suitability for carbon sequestration. These sites include oil and gas fields, depleted or under production, as well as saline aquifers, belonging to the territorial sea of the United Kingdom, Norway, Denmark, the Netherlands and Germany. Where possible we compare with previous storage estimates in the literature.

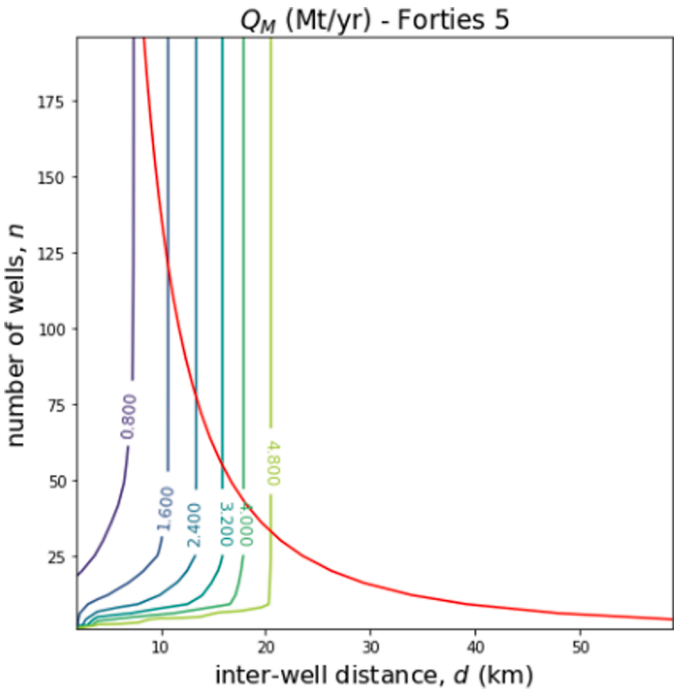



**Fig. 2.** and the associated text shows plausible $CO_2$ flowrate scenarios for the Forties oil field: this is a target rate in units of mass per unit time. Coloured lines represent the different scenarios of $CO_2$ flowrate, while plausible scenarios are those to the left of the red curve. Post injection or monitoring periods, though important in real conditions, are not included in this preliminary study. We assume that all the injected mass is retained in the subsurface.

## 2. Methodology

The tool used to estimate storage capacity is the CO2BLOCK, open source, software developed by De Simone and Krevor (2021) (available here) that receives as input the geophysical characteristics of specific fields to estimate the capacity as a function of the number of injection wells. The model considers only physical processes occurring in subsurface; no chemical processes are considered. The software uses a sophisticated analytical model of pressure dissipation – which has been validated against numerical solutions (De Simone and Krevor (2021) and references therein). The model distinguishes between open and closed type aquifers. As mentioned in the introduction, assessments of storage capacity have been performed for UK's oil and gas fields as well as saline aquifers. The most extensive study is reported in the

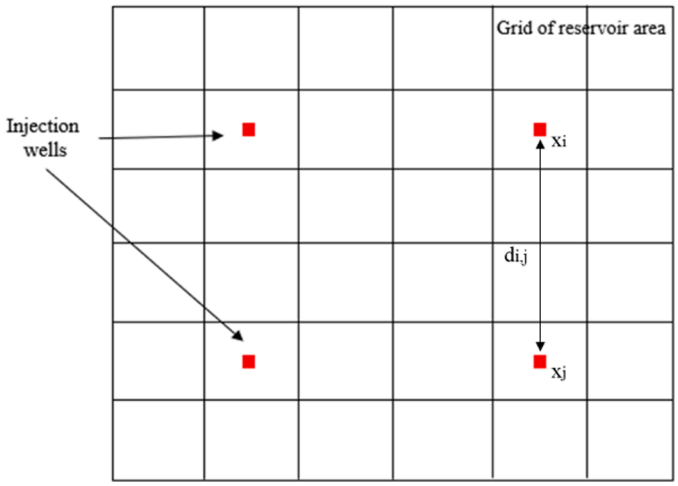



**Fig. 3.** Grid of reservoir area and injection well placement.

CO2STORED database (CO2STORED 2021) including 500 storage units within Irish and North Sea and Central English Channel (Bentham et al., 2014). The results provide, among others, theoretical capacity, injection rate, number of wells and storage efficiency. In the case of saline aquifers, the accessible pore volume is calculated as the product of available pore volume, of compressibility and limiting pressure for storage. The latter is set to be equal to the 90% of lithostatic pressure. This constraint is not considered for open aquifers through which $CO_2$ migrates. Gammer et al. (2011) developed a storage calculation methodology for oil and gas fields where the pore volume available for $CO_2$ storage was equated to the net volume of fluids extracted during hydrocarbon exploitation. For saline aquifers the estimation of capacity was based on pore volume available considering or not hydraulic isolation. In all cases, an overall pressure limit that constraints capacity is assumed. In addition several previous studies have estimated capacity by considering the gross rock volume of the formation only, usually producing over-estimates since the transient pressure response is ignored (see, for instance, (Thibeau et al., 2014, Cook and Zakkour, 2015, Goodman et al. (2014), Tasianas et al. (2016))). Our analysis is distinct in that it accounts for injection rate and limits the pressure reached during the injection phase which tends to lead to a more severe constraint on storage capacity. The effects of pressure dissipation on constraining storage capacity are supported by Szulczewski et al. (2014) (Szulczewski et al., 2014b)who showed that for low permeability, high porosity aquifers, pressure limited capacity tends to be very important. Hence, the model utilized in this study, is an improvement on simple volume or pressure-limited assessments. It employs a semi-analytical calculation of the pressure response to $CO_2$ injection, first introduced by Nordbotten et al. (2005) while the analysis is based on De Simone et al. (2019) who developed a methodology that maximizes storage respecting the pore pressure limit in each reservoir.

Fig. 1 demonstrates the workflow of the model to estimate the storage capacity.

### 2.1. Pressure response to $CO_2$ injection

In this study, the generalized solution derived by Nordbotten et al. (2005) was used to calculate pressure build-up due to $CO_2$ injection into a single well in a reservoir with open boundaries and under the assumption of no fractures and traps. The latter is calculated as:

$$\Delta p(r,t) = \frac{Q\mu_w}{2\pi KH} \times \begin{cases} \frac{\mu_c}{\mu_w}\ln\left(\frac{\psi}{r}\right) + \ln\left(\frac{R}{\psi}\right), & r < \psi \\ \ln\left(\frac{R}{r}\right), & \psi < r < R \\ 0 & r > R \end{cases} \quad (1)$$

where $\Delta p(r,t)$ is the pressure variation at time $t$ and distance $r$ from the injection well, $Q$ is the volumetric flow rate, $K$ is the absolute permeability, $H$ the reservoir thickness, $\mu_w$ and $\mu_c$ are brine and $CO_2$ viscosities respectively, $c$ is the reservoir compressibility, while, $\psi$ represents the radius of a fictious equivalent vertical interface such that $\psi = \exp(\omega)\xi$ and $\omega = \frac{\mu_c+\mu_w}{(\mu_c-\mu_w)}\ln\left(\sqrt{\frac{\mu_c}{\mu_w}}\right) - 1$, $\xi = \sqrt{\frac{Qt}{\pi\phi cH}}$, $\phi$ is the porosity, $R$ refers to the radius of pressure propagation that grows proportionally to $\sqrt{t}$ and is determined as:

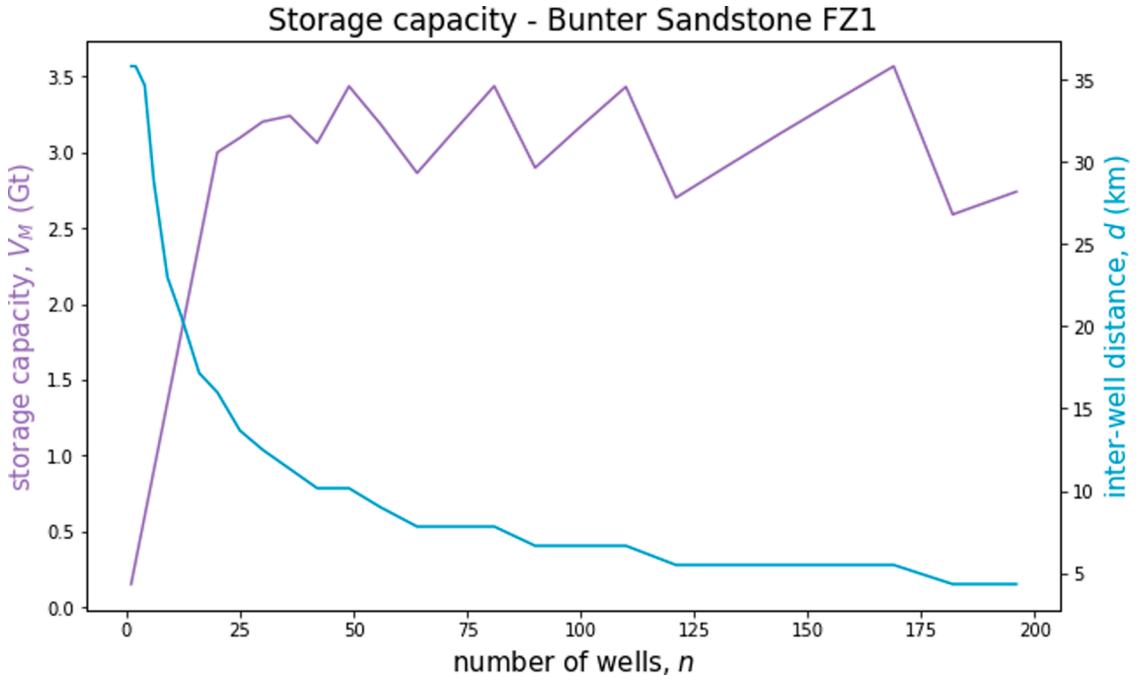



**Fig. 4.** Storage capacity as a function of wells and inter-well distance for Bunter sandstone formation zone 1.

**Table 1**
CO2BLOCK required input parameters.

| Parameter | Default value | Reference |
|---|---|---|
| Aquifer type (Open/Closed) | specific to field | – |
| Shallowest depth [m] | specific to field | – |
| Mean depth [m] | specific to field | – |
| Reservoir thickness [m] | specific to field | – |
| Field Area [$m^2$] | specific to field | – |
| Permeability [mD] | specific to field | – |
| Porosity [-] | specific to field | – |
| $CO_2$ density [tonne $m^{-3}$] | 0.662 | Redlich and Kwong (1949) |
| $CO_2$ viscosity [mPa.s] | 0.0146 | Altunin et al. (1972) |
| Brine viscosity [mPa.s] | 1.0518 | (Batzle and Wang, 1992) |
| Pore pressure at shallowest depth [MPa] | specific to field | – |
| Mean pressure [MPa] | specific to field | – |
| Salinity [ppm] | 89,000 | De Simone et al. (2021) |

$$R = \sqrt{\frac{2.25Kt}{\mu_w c}} \tag{2}$$

The factor of *2.25* comes from the approximation of the well function proposed by Cooper and Jacob (1946). In the case of a closed reservoir, under the assumption that $CO_2$ plume radial extension $\psi$ is smaller than reservoir radius $R_c$, the above equation is modified as:

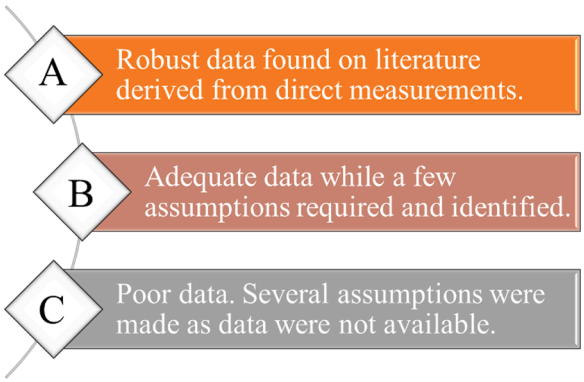



**Fig. 6.** Data classification based on the robustness of the data used.

**Table 2**
Default parameters that are used if they are not provided.

| Parameter | Value |
|---|---|
| Lithostatic gradient | 23 MPa $km^{-1}$ |
| Hydrostatic gradient | 10 MPa $km^{-1}$ |
| Temperature gradient | 33 °C $km^{-1}$ |
| Default stress ratio ($s_3/s_1$) | 0.7 |
| Rock cohesion | 0 MPa |
| Rock compressibility | $5 \times 10^{-4}$ $MPa^{-1}$ |
| Water compressibility | $3 \times 10^{-4}$ $MPa^{-1}$ |

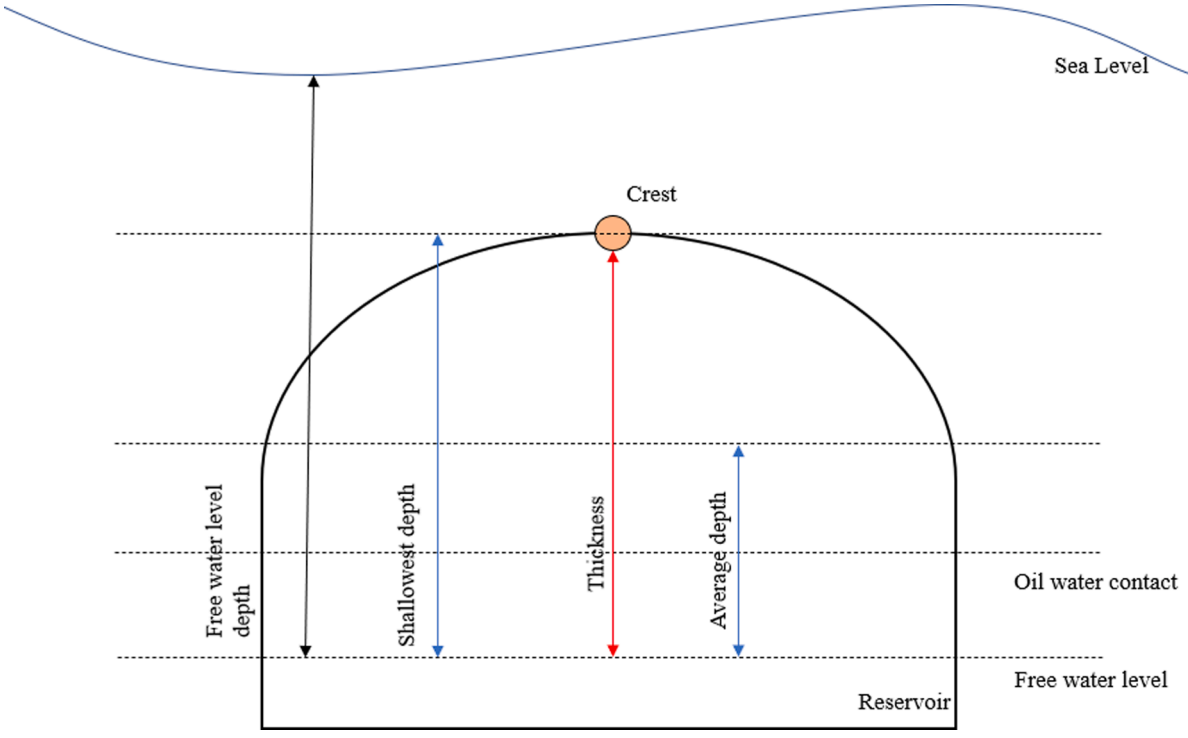



**Fig. 5.** How reservoir depths were calculated when direct data were absent.

**Table 3**
Characterization of data sources and relevant assumptions.

| Country | Data score | |
|---|---|---|
| United Kingdom | **A** | Some assumptions only in Cormorant, Nansen, Bunter and Statfjord saline aquifers regarding the depth. |
| Denmark | **A** | Some assumptions in 3 oil fields regarding their area. |
| Norway | **B** | Assumptions regarding the volume of the saline aquifers. |
| Germany | **A** | No extra assumptions. |
| Netherlands | **C** | Assumptions regarding the area of the oil and gas fields, while all other geophysical and geospatial characteristics were provided in the literature. Therefore, an area of 76 $km^2$ per field was assumed as an average of the oil and gas fields of the other great database available, the one for the UK. |

**Table 4**
Databases used in the analysis.

| Country | Reference |
|---|---|
| United Kingdom | ■ CO2STORED<br>■ (Gluyas and Hichens, 2003)<br>■ Zimmerman (1990) |
| Netherlands | ■ Netherlands Geological Survey (https://www.nlog.nl/) (Netherlands Geological Society 2021) |
| Norway | ■ Norwegian Petroleum Directorate (https://factpages.npd.no/) (https://factpages.npd.no/en 2021a)(https://factpages.npd.no/en 2021b)<br>■ (Halland et al., 2017) |
| Denmark | ■ Danish Energy Agency (https://ens.dk/)(https://ens.dk/en 2021)<br>■ Hardman (1982) |
| Germany | ■ (Grassmann et al., 2005)<br>■ Anthonsen et al. (2014) |

$$\Delta p(r,t) = \frac{Q\mu_w}{2\pi kH} \times \begin{cases} \frac{\mu_c}{\mu_w}\ln\left(\frac{\psi}{r}\right) + \ln\left(\frac{R_c}{\psi}\right) + \frac{2R^2}{2.25R_c^2} - \frac{3}{4}, & r < \psi \\ \ln\left(\frac{R}{r}\right) + \frac{2R^2}{2.25R_c^2} - \frac{3}{4}, & \psi < r < R \\ 0 & r > R \end{cases} \quad (3)$$

For multi-well injection, the pressure response at location $x_i$ can be approximated as the superposition of single well pressure responses

$$\Delta p_{sup}(x_i,t) = \sum_{j=1}^{n} \Delta p(d_{ij},t) \quad (4)$$

where $n$ is the number of wells, $\Delta p(r, t)$ is the pressure response to a single well injection at distance $r$ from the injection well, while $d_{ij}$ is the distance between location $x_i$ and $x_j$. Assuming open boundaries with the same flowrate at each well the superposed pressure response at $x_i$ is estimated as

$$\Delta p_{sup}(x_i,t) = \frac{Q\mu_w}{2\pi kH}\left[\frac{\mu_c}{\mu_w}\ln\left(\frac{\psi}{r_0}\right) + ln\left(\frac{R_c}{\psi}\right) + \sum_{j=2}^{n}\ln\left(\frac{R}{d_{ij}}\right)\right] \quad (5)$$

### 2.2. Δp/Q relationship

The extension of the $CO_2$ plume is affected by the flowrate. This implies that Δp does not have a linear dependence on $Q$ for $r < \psi$. For both open and closed reservoirs, the difference between the over-pressure associated with two different flowrates is calculated as:

$$\Delta p_D(r,\ t,\ Q_2) - \Delta p_D(r,\ t,\ Q_1) = -\delta\ln\left(\frac{\psi_2}{\psi_1}\right) = -\frac{\delta}{2}\ln\left(\frac{Q_2}{Q_1}\right) \quad (6)$$

and

$$\delta = \frac{\mu_w - \mu_c}{\mu_w} \quad (7)$$

rearranging the previous equation,

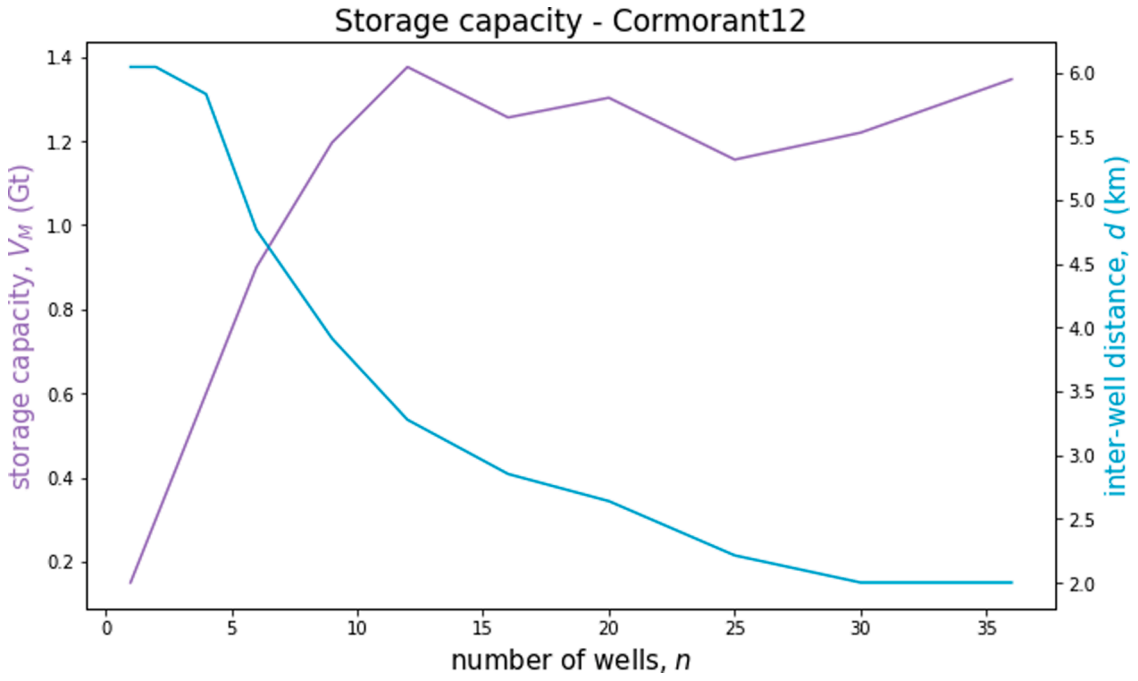



**Fig. 7.** Storage capacity as a function of well number and inter-well distance for the Cormorant formation.

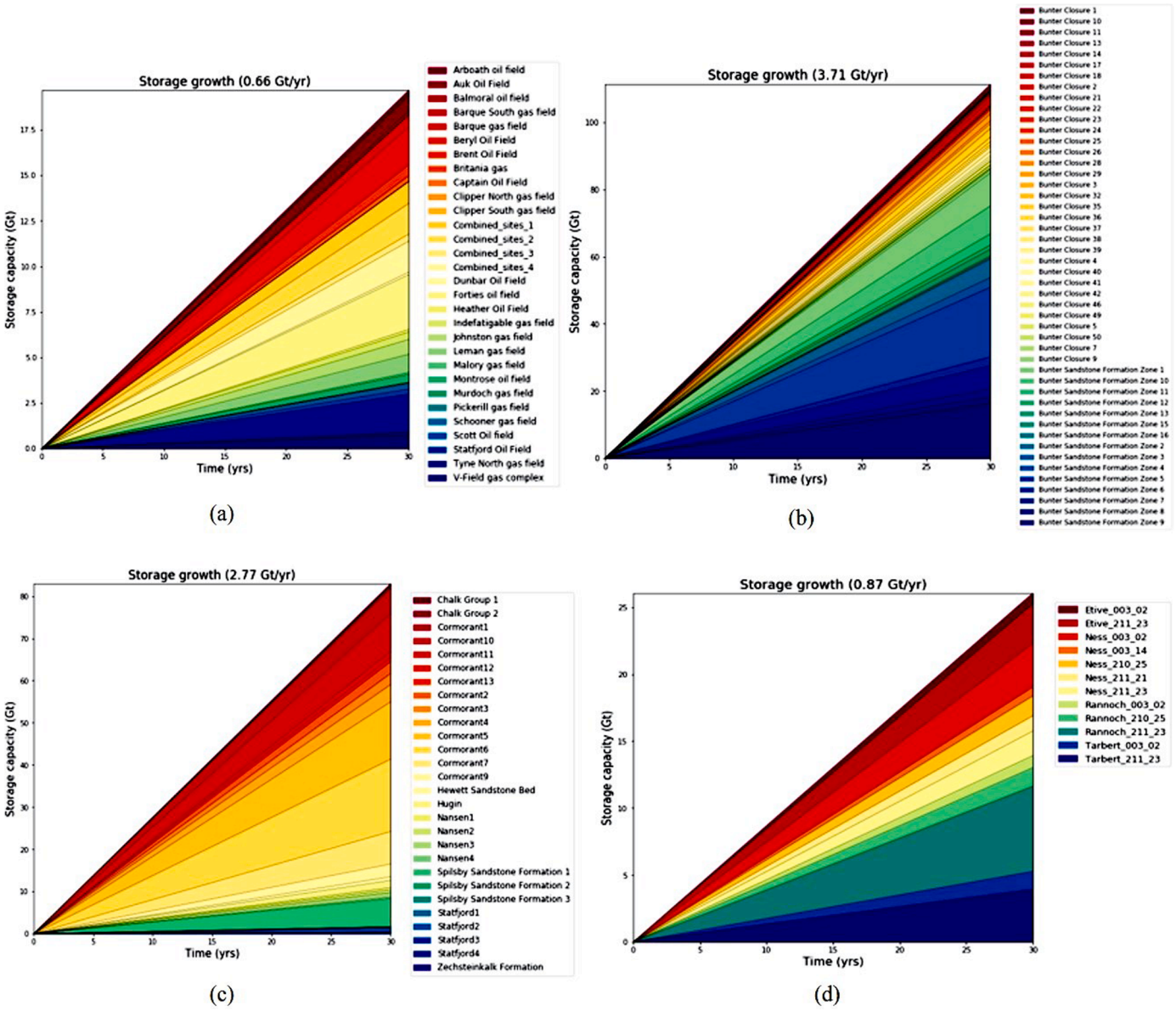



**Fig. 8.** Storage capacity in the United Kingdom's oil & gas fields (a), Bunter sandstone formations (b), various other formations (c), and Brent sandstone formations.

$$\Delta p(r,t,Q_2) = Q_2\left[\frac{\Delta p(r,t,Q_1)}{Q_1} - bln\left(\frac{Q_2}{Q_1}\right)\right] \tag{8}$$

and

$$b = \frac{\mu_w - \mu_c}{4\pi kH} \tag{9}$$

Solving the above equation for $Q_2$ results in the calculation of flowrate after having estimated the overpressure for the first case $Q_1$.

### 2.3. Limits to pressure build-up

#### 2.3.1. Maximum flowrate

The next step is the estimation of plausible scenarios of the inter-well distance and number of wells, for a given injection rate in the reservoir: this is a target reservoir volume per unit time, converted to injected mass per unit time. We assume that all the injected mass is retained in the subsurface. In this way, the maximum plausible injection rate, subject to constraints, can be evaluated. Pressure build-up in the reservoir decreases in a non-linear way, as well as depending on the number of wells and their spacing. Therefore, a number of potential designs of $CO_2$ injectivity, respecting the pressure limit threshold $\Delta P_M$, are produced. Once the maximum allowable pressure has been estimated, we can use it to determine the maximum sustainable flowrate $Q_M$. This rate leads to the estimation of the maximum amount of $CO_2$ that can be stored in a closed time interval $t$, without exceeding the suggested pressure limit. In single-phase flow, pressure build-up and injection flowrate increase linearly, while in multiphase flow (Szulczewski et al., 2012) the same relationship can be assumed for small variations of flowrate. This allows the estimation of the maximum allowable injection flowrate as:

$$Q_M(t) = \frac{Q_r \Delta p_M}{\Delta p_r(t)} \tag{10}$$

where $\Delta P_r(t)$ is the pressure response for reference flowrate $Q_r$. The direct estimation provides an assessment of the maximum flowrate for a number of scenarios, illustrated in Fig. 2.

### 2.4. Constraints of the model

The first constraint has to do with the need to avoid $CO_2$ plume

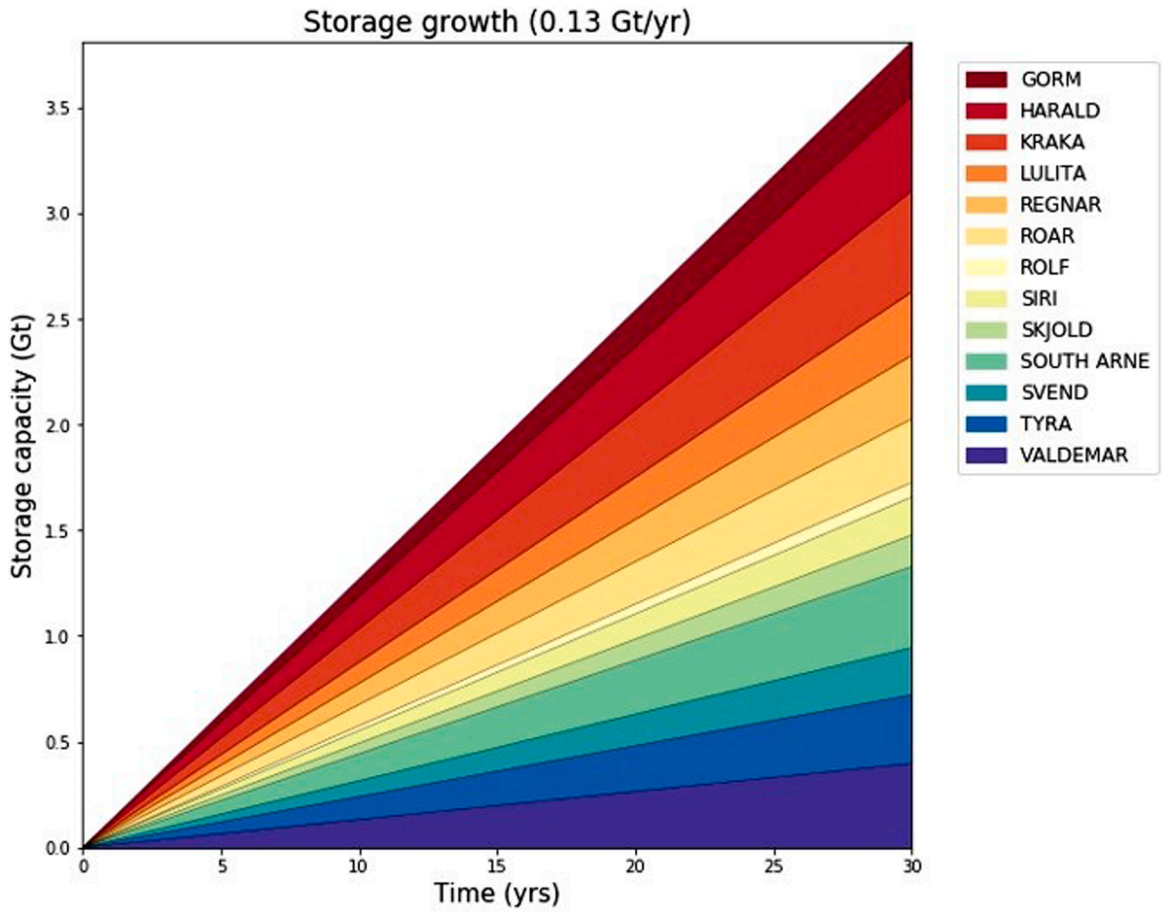



**Fig. 9.** Storage capacity in fourteen oil and gas fields in of Denmark.

interference. In this constraint, the half of the inter-well distance is smaller than the plume average propagation distance resulting in

$$d > 2\sqrt{\frac{Q_M t}{n\pi\phi \mathrm{H}}} \tag{11}$$

Another constraint limits the number of wells in a given area (referring to the area of the reservoir $A$). Assuming that the well distribution is a Cartesian grid (Fig. 3), the constraints imposed by the area size of the reservoir is given by

$$d \leq \sqrt{\frac{A}{n}} \tag{12}$$

In the case of a closed aquifer, the available recovery area is reduced according to the following equation to approximate the reduced capacity:

$$r_c = \sqrt{\frac{\mathrm{A}}{\pi}} \tag{13}$$

Apart from these constraints, there are technical limitations on the maximum injectable flowrate per well given by current technology, $Q_s$, such that

$$n \geq \frac{Q_M^{tot}}{Q_s} \tag{14}$$

where $Q_M^{tot}$ is the total injection rate referring to the sum of the injection of the $n$ wells. Results for the storage capacity are reported in Fig. 4 as a function of the number of wells and inter-well distance.

### 2.5. Assumptions

The analytical model used has been extensively validated (Nordbotten et al., 2005; Szulczewski et al., 2012). We assume that capacity is limited by the build-up of pressure and that all the $CO_2$ injected remains underground. We do not account for long-term trapping processes, such as capillary trapping, dissolution and reaction which may allow higher trapping capacities to be achieved. Table 1 presents the software parameters that need to be estimated, retrieved from log (downhole) measurements or calculated along with some default values in case some of those data are not accessible. In addition, the model utilized, considers virgin oil and gas fields without taking into account previous injection or production of hydrocarbon, water or other fluids.

### 2.6. Depth

Whenever data for shallowest depth, average depth and thickness were not directly available the following were considered. Shallowest depth was calculated as the difference between the free water level and thickness. While average depth is calculated as free water level depth minus half thickness. When free water level depth data were not available, instead, oil water contact was used for the calculation as shown in Fig. 5.

### 2.7. Area

For area, when adequate data were not available, two approaches were followed in this project. Providing that values for gross rock volume exist, the following equation used:

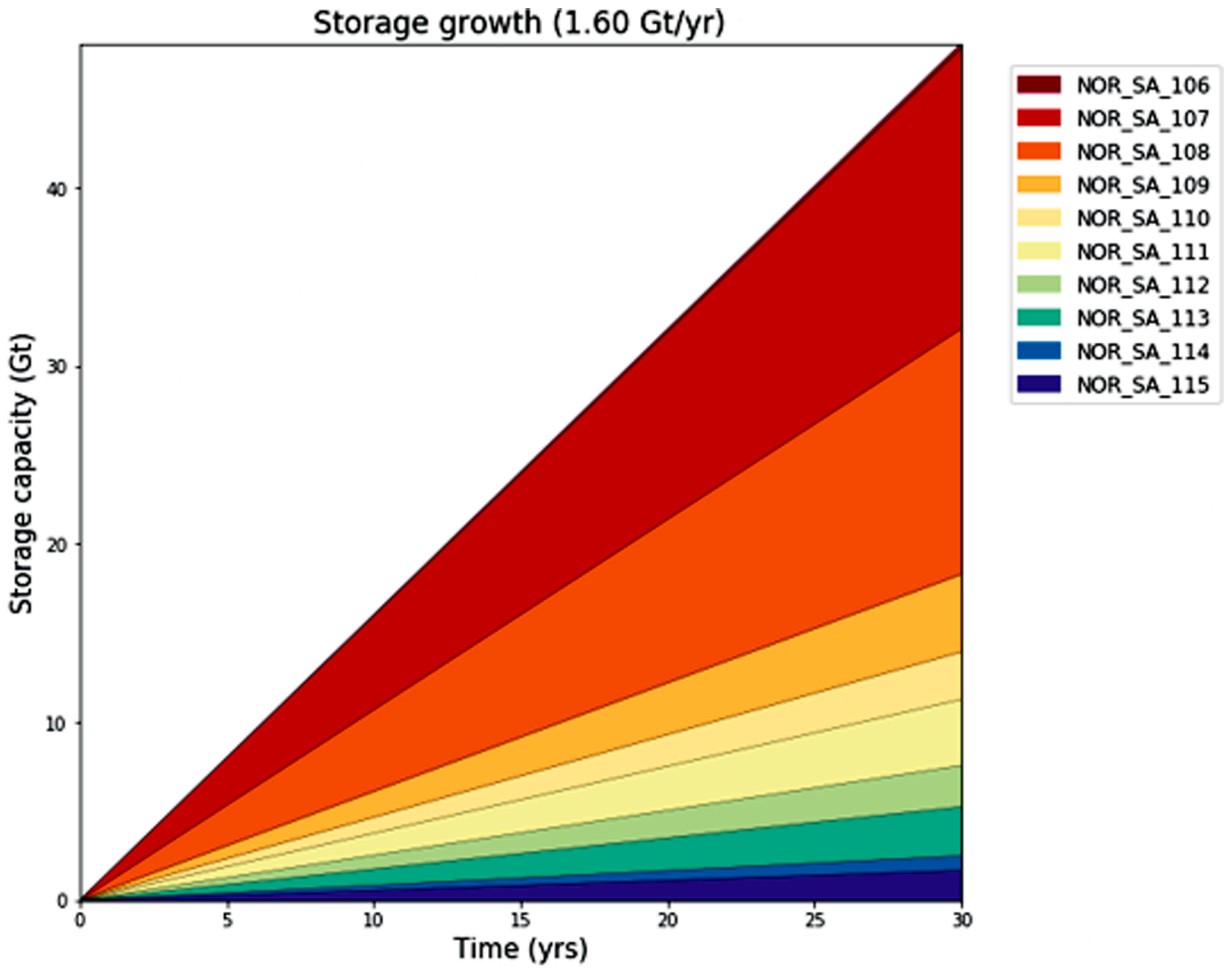



**Fig. 10.** Storage capacity in Norway's saline aquifers.

$$A = \frac{V}{H} \tag{15}$$

where $V$ is the gross rock volume and $H$ is gross thickness (m).

Otherwise, gross rock volume is calculated from stock tank oil initially in place (STOIIP) with Eq. (16) to calculate area:

$$STOIIP = \frac{V \times \phi \times S_o}{B_o} \tag{16}$$

where $V$ is the gross rock volume, $\phi$ is porosity (fraction), $S_o$ is residual oil saturation, $B_0$ is oil formation volume factor (the conversion of reservoir to surface volume). When, for an oil field, STOIIP is not known, the area is estimated from available GIS photographs. Especially, for the area of the Netherlands oil and gas fields, where data were not available online, we used an approximation. The average area of oil and gas fields in the UK (where data are available) is calculated and the result is used for every field in the Netherlands. Although highly uncertain, this approximation enables a first order estimation of a total capacity of carbon dioxide storage in the Netherlands.

### 2.8. Porosity and permeability

In a reservoir or a saline aquifer, porosity and permeability differ across space and log (down-hole) measurements provide a range of results. Usually, in databases, average values of permeability or porosity are reported. In other cases, a range of log measurements results are provided. In the later, the average is used in the calculations. When data for porosity and permeability were not available, average values corresponding to similar geological formations, gleaned from the literature, are used.

### 2.9. Data characterization

The properties introduced above are characteristics unique for each field, and are found from down-hole (log) measurements and the analysis of rock samples taken from the reservoir. In most cases, these data are available from independent studies and are reported in the literature, but in other cases they need to be estimated under the assumptions presented above. An index on the data was created ranging from A to C (robust to poor data), shown in Fig. 6, while the essential assumptions are discussed in Table 3.

In the case that some parameters are not provided in data sources or in literature, further approximations are needed as listed below.

(i) *Pressure at the top of the reservoir*

$$P_0 = Hydrostatic\ gradient\ \times mean\ depth$$

(ii) *Temperature at the center of the reservoir*

$$T_0 = Temperature\ gradient\ \times depth$$

(iii) *Rock compressibility*

$$c_{rock} = c_{default}$$

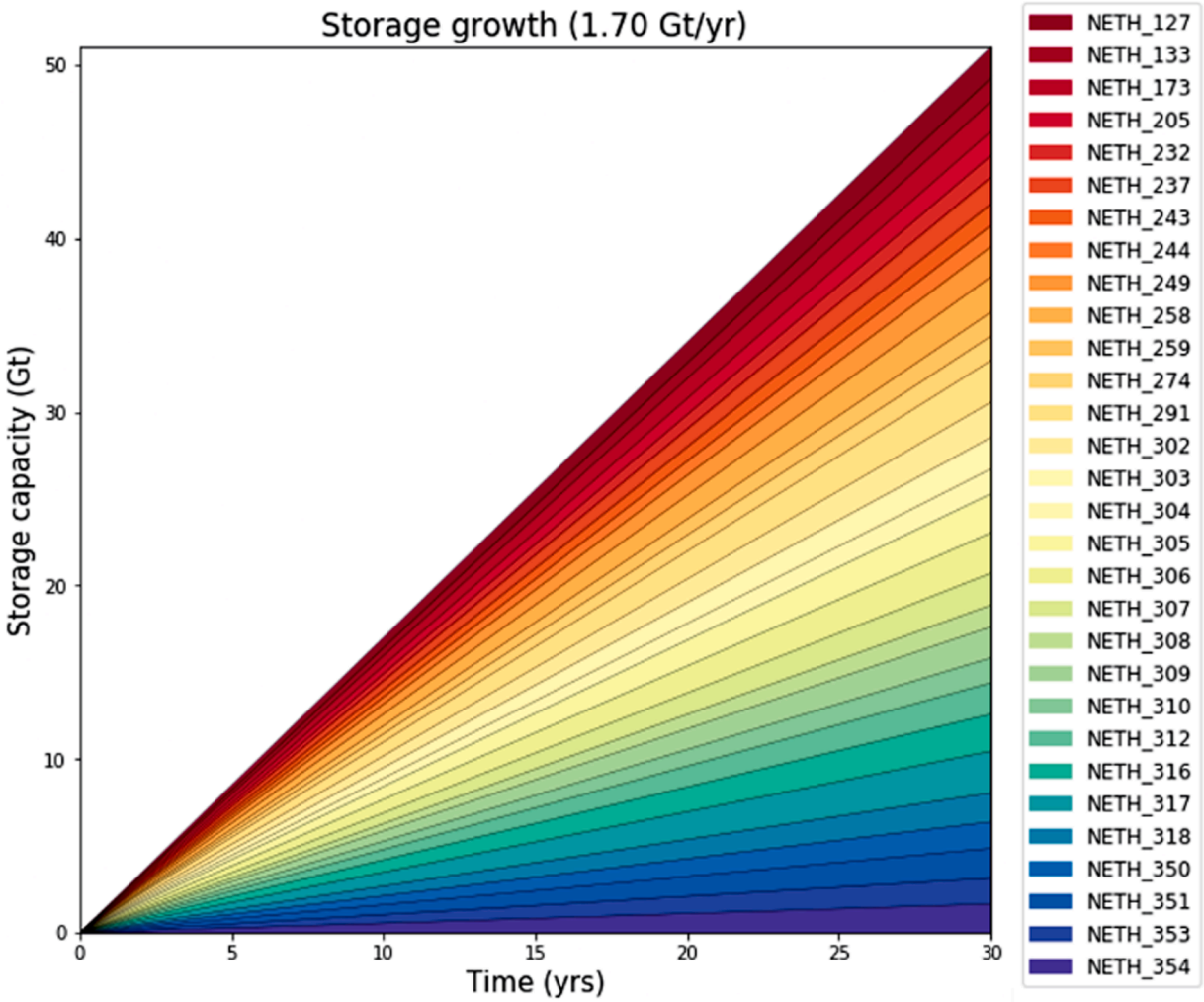



**Fig. 11.** Storage capacity in the thirty largest oil and gas fields in the Netherlands.

Several default parameters are also used in case they are not provided and are reported in Table 2.

A sensitivity analysis on the software showed only a small dependence of predicted capacity on salinity or temperature gradient, and hence these values were considered constant. Stress ratio, water and rock compressibility have a quasi-linear correlation on ultimate capacity; the average values demonstrated in Table 2, were used. Finally, and due to lack of availability, in all the fields examined, non-cohesive rocks were considered.

### 2.10. Storage efficiency

The storage efficiency quantifies how much void space is occupied by carbon dioxide, with respect to the available pore volume of the rock. The volume of $CO_2$ depends on its density $\rho_{CO2}$, which is a function of temperature and pressure (Bachu 2015):

$$E = \frac{M_{CO_2}}{\rho_{CO_2} \times A \times H \times \phi} \tag{17}$$

where $M_{CO_2}$ is the carbon dioxide mass stored in the aquifer which has been calculated previously, $A$ is the reservoir area, $H$ refers to the reservoir thickness and $\phi$ is the porosity of the rock.

According to these equations, apart from the stored mass of $CO_2$ that is a function of a set of parameters analyzed in the previous chapter, storage efficiency is affected by the designated area of injection ($A$) and the porosity of the rock ($\phi$). For the density of the injected carbon dioxide at surface conditions, a fixed value was chosen to be 0.662 tonne $m^{-3}$.

## 3. Results

In this section, the North Sea $CO_2$ storage capacity potential is reported, following the methodology described previously, for territories belonging to the United Kingdom (UK), Denmark, Norway, the Netherlands and Germany. Several offshore oil and gas fields were considered as well as saline aquifers. The fields studied in every country were chosen dependent on data availability. For the United Kingdom, 82 oil and gas fields were considered (some of them depleted or close to depletion, while others are fully operating) and 88 saline aquifers (12 Brent formations, 48 Bunter Sandstone formations, 13 Cormorant fields, 4 Statfjord formations, 4 Nansen, 3 Spilsby Sandstone formations, and 2 Chalk Group formations). For Denmark, adequate data were retrieved for 13 oil and gas fields, while for Norway 10 deep saline aquifers were considered. In Germany, a large oil field (Mittelplate) was studied along with two Bunter formation aquifers. For the Netherlands 247 oil and gas fields (depleted or not) were considered due to huge databases available online. The databases used in this study are provided in Table 4. More fields in the territory of the North Sea could potentially sustain carbon dioxide injection and storage process but a lack of geophysical data

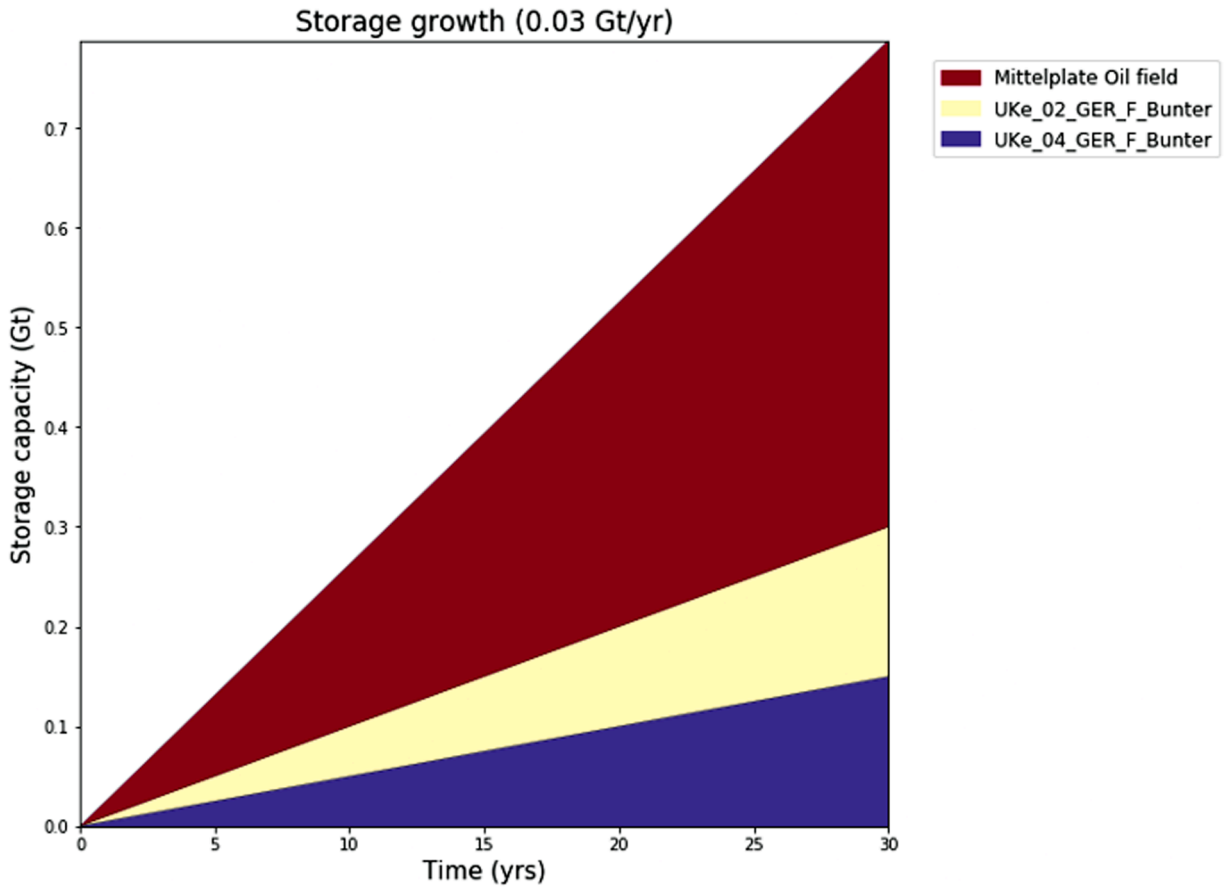



**Fig. 12.** Storage capacity in Germany's Mittelplante oil field and saline aquifers in the Bunter sandstone formation.

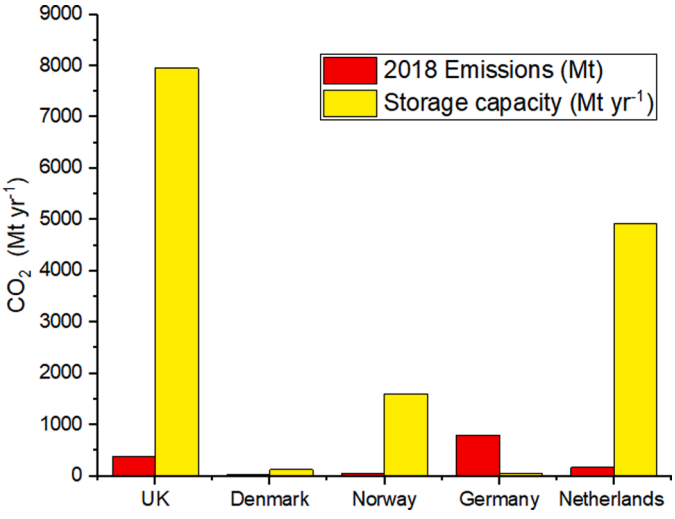



**Fig. 13.** Carbon dioxide emissions per year and storage potential in the formations studied.

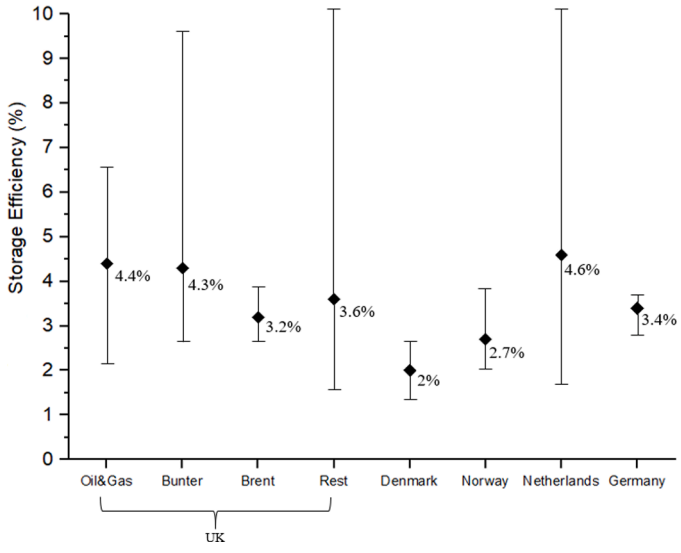



**Fig. 14.** Average storage efficiency in the selected hydrocarbon fields and saline aquifers.

excluded them from this study. Apart from these, as this paper includes fields that have been studied before using other methodologies, it provides another approach for estimating capacity.

### 3.1. United kingdom

In the United Kingdom's oil and gas fields, a total of 18.2 Gt of $CO_2$ can be stored over 30 years, employing around 195 wells. The greatest capacity is observed (Fig. 8, (a)) for the Forties oil field, a massive 93 $km^2$ field characterized by high permeability and porosity (Carman et al., 1981) that alone can store 3.0 Gt of $CO_2$. Also, the Leman and Brent oil fields are among of those which capacity exceeds 1.0 Gt in over 30 years. Bunter sandstone formations (Fig. 8, (b)) are characterized by permeabilities on average, close to 100mD and low porosities around 12%. On the other hand, their massive volume resulted in large

**Table 5**
Results of storage capacity in the countries considered.

| Country | Total $CO_2$ storage capacity (Gt) | $CO_2$ injection rate (Mt $yr^{-1}$) |
|---|---|---|
| United Kingdom | 238 | 7954 |
| *UK Oil & Gas fields* | 18.2 | 608 |
| *UK Brent formations* | 26 | 869 |
| *UK Bunter formations* | 111 | 3710 |
| *Other formations in UK* | 82 | 2765 |
| Denmark | 3.8 | 127 |
| Norway | 48 | 1603 |
| Netherlands | 147 | 4925 |
| Germany | 1.64 | 55 |
| **Total** | **440** | **22,620** |

capacities, as the 48 saline aquifers considered, resulted in 110 Gt of $CO_2$ storage employing 1340 wells. From the various formations studied, the Cormorant fields provided large capacities, while all the saline aquifers of the various formations, can store more than 80 Gt using 1050 wells. In the Brent sandstone formations, saline aquifers, (Fig. 8(d)) 26 Gt of $CO_2$ can be stored cumulatively in a 30-year period employing more than 300 wells. From the latter, Rannoch, a highly permeable aquifer has a total potential of 6.3 Gt followed by Ness and Tarbert with capacities exceeding 3 Gt. It is important to note that storage capacity is a function of the number of wells drilled in a bounded area. Following Fig. 7, we observe that capacity increases with the number of wells, until a plateau is reached, beyond which capacity remains constant. Indeed, since the capacity calculation approach is pressure driven, a great number of wells, may also penalize storage due to an excessive pressure increase.

### 3.2. Denmark

For Denmark's oil and gas fields that were examined, we estimated 3.8 Gt of $CO_2$ in a 30-year time frame (Fig. 9). The fields considered belong to low permeability and high porosity formations, with medium to low gross rock volumes. Nevertheless, as explained below, they are sufficient to offset the country's emissions.

### 3.3. Norway

In the ten saline aquifers studied in Norway, a total capacity of 50 Gt of $CO_2$ can be potentially stored, at a rate of 1.6 Gt per year (Fig. 10). The greatest capacity is reported for the Utsira and Skade aquifers (NOR_107) with a storage potential of 15.7 Gt, while the Bryne and Sanders formation (NOR_108) is capable of storing close to 14 Gt over 30 years. All the selected aquifers are able to store more than 2 Gt of $CO_2$ and are characterized by enormous gross rock volumes, high porosities and medium to high permeabilities.

### 3.4. The Netherlands

From the total 247 oil and gas fields considered in the Netherlands, the thirty with the greatest capacity are shown in Fig. 11. In total, 1440 injection wells, are able to inject close to 150 Gt of $CO_2$ in 30 years providing an enormous storage potential. The geophysical characteristics have a wide variation, but on average, such reservoirs possess, low porosity with an average of only 0.13, AND HIGH PERMEABILITY CLOSE TO 500 MD.

### 3.5. Germany

Germany is a country characterized by few offshore oil and gas fields, with the Mittelplate oil field being the largest in size and storage potential (Fig. 12). It is estimated that with a total of 6 wells, it can store around 630 Mt of $CO_2$ over 30 years. In addition, two saline aquifers of the Bunter sandstone formation, increase the previous amount to 1.65 Gt, with a 0.03 Gt per year injection rate.

### 3.6. Potential capacity vs demand for CCS and comparison with previous studies

A very important outcome is to compare storage capacity against projected emissions: this is presented graphically in Fig. 13. The United Kingdom has the potential to store 8 Gt per year, a value twenty times higher than its current emissions (Blasing, 2016). The Netherlands, with a maximum potential injection rate close to 5 Gt per year can offset 30

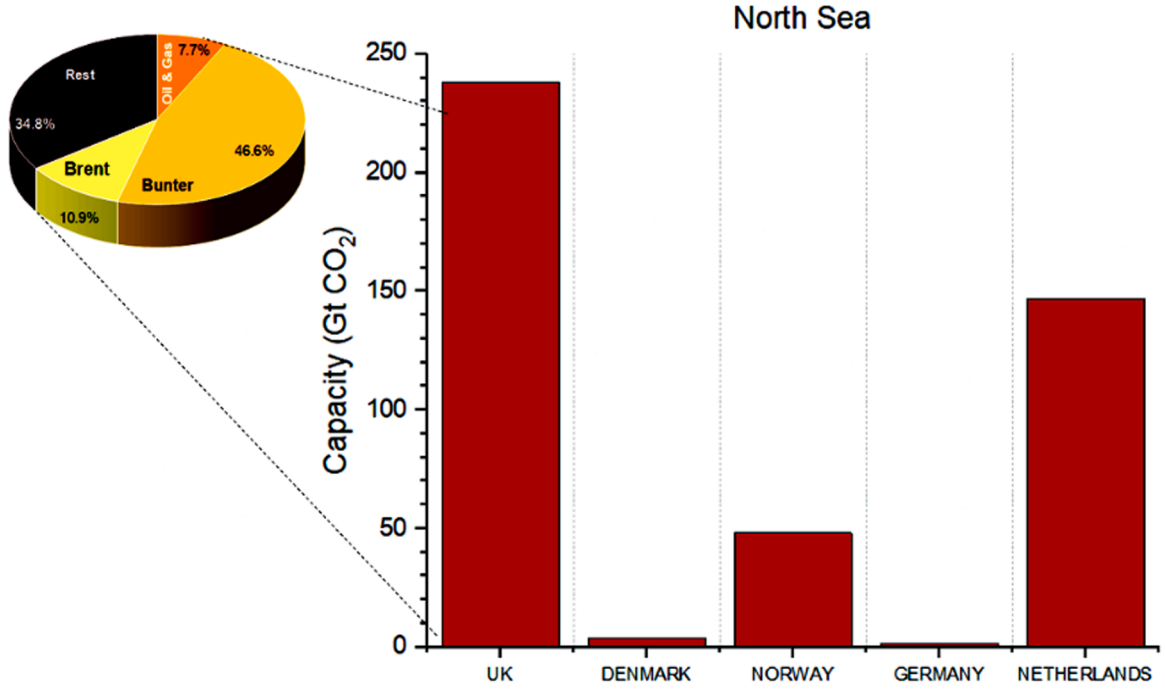



**Fig. 15.** Cumulative results on carbon dioxide storage in North Sea territories as calculated in this study.

times its current $CO_2$ emissions (Ortiz et al., 2020) while Norway, has a potential of 1.6 Gt per year, an amount 36 times higher than its annual upstream $CO_2$ emissions. It is worth clarifying, that this amount refers only to saline aquifers and does not take into account any oil and gas fields. Denmark can store underground, in the selected oil and gas fields, more than two times its annual emissions reducing significantly its industrial environmental footprint. In Germany, the selected fields can store $CO_2$ with an annual rate of 0.03 Gt an amount not comparable with current emissions; in this case spare capacity from neighboring countries could be used. It is important to mention at this point, that the history of oil and gas fields are not taken into consideration as this is a theoretical study considering only geophysical properties of the rocks and not the production drive mechanisms specific to these fields.

Our estimates can also be compared with previous studies in the literature. In most cases, deviations in the results are the outcome of different calculation approaches or, most commonly, the different assumptions that authors consider to estimate missing data. Anthonsen et al. (2014) studied exactly the same oil and gas fields in Denmark, using a volume-based approach, that resulted in an ultimate capacity of 2.2 Gt, a value close to the 3.8 Gt estimated in this study. Halland et al. (2017) conducted a study on Norwegian saline aquifers, utilizing the gross rock volume approach that resulted in 44.8 Gt of potential $CO_2$ storage capacity, very similar to the 48 Gt calculated in this paper. In the United Kingdom, several studies have been performed, including De Simone et al. (2019) who examined 25 oil and gas fields as well as several Bunter and Brent formations that resulted in 140 Gt $CO_2$ of capacity while this investigation resulted in a total of 238 Gt $CO_2$. Bentham (2006) conducted a study on several oil and gas fields resulting in 5.6 Gt $CO_2$ stored, in approximately half of the fields included in this study and under this methodology that resulted in around 18.2 Gt. A study by the Energy Technology Institute (Gomersall and Murphy, 2016) reported an estimated 0.28 Gt of $CO_2$ capacity in Bunter Closure 36 considering a 40-year injection period and 7 Mt/y injection rate derived from neighboring industrial processes. In this paper, the capacity was estimated at 1.29 Gt in a time span of 30 years under 3.59 Mt/y per well, for 12 wells, injection rate of $CO_2$; this is a considerably larger volume because our model allows a higher injection rate spread over multiple wells and is not limited by the availability of $CO_2$. Both this study and our paper examines Captain field and Forties. On the other hand, the study of Energy Technology Institute examines only a specific field sites (Captain site X and Forties 5 site 1) which makes it impossible to compare with our results, as we have included all the geographical area of such sites in volume estimation and eventual storage evaluation.

### 3.7. Storage efficiency

Fig. 14 is a graphical representation of the results obtained for the average storage efficiency in the fields considered in the study. Oil and gas fields in the UK present an average efficiency close to 4.5% (2.1% - 6.6%), while for Bunter formations a wider range is noted with the average being 4.3% (2.2% - 9.5%). Brent formations result in a lower efficiency close to 3.2% due to higher porosities and greater areas, values that are denominators in the ratio in Eq. (17). In the other nations studied, the oil and gas fields in Denmark present an efficiency of 2% (1.7% - 2.5%) as they are characterized by higher permeability and smaller surface area than in the UK. Norway's saline aquifers are characterized by efficiencies around 2.7% (2.2% - 3.7%) due to great rock volumes (Sanders formation 6 $km^2$, Ustria formation 2.8 $km^2$). The large number of oil and gas fields in the Netherlands present a range of storage efficiency from 2% to 10% with an average of 4.6%, while Germany's fields present medium sized gross rock volumes with an average efficiency of 3.4% (3.3% −3.7%). These numbers appear small, but the volumes of the storage aquifers are huge, and so still provide more than adequate capacity as discussed previously.

## 4. Conclusions

Carbon capture and storage is a necessary mitigation strategy to avoid dangerous climate change. Extensive $CO_2$ capture and injection into the subsurface prevents its release into the atmosphere and stores it permanently for several thousand years. In Europe, the greatest part of its carbon dioxide emissions is concentrated in the boundary areas/regions of the North Sea and are a product of substantial industrial activity in the countries of the north.

This study is a comprehensive analysis of 441 potential locations in the North Sea territory that can be used as carbon storage sites, categorized based on the countries of origin. These are oil and gas fields, depleted or not, and several saline aquifers, with the most common belonging to the Bunter and Brent formations. Though other studies have examined the storage capacity in specific saline aquifers and/or oil and gas fields, the absence of an inclusive study for the North Sea territory was identified. This was the motivation for this comprehensive case study, taking into account the transient pressure response upon injection, which allows us to provide a first order estimation of storage availability. The work can be used for further development of national strategy and cost analysis pertaining to $CO_2$ storage. It has been found that over 30 years, a total amount of 440 Gt of $CO_2$ can be stored in the fields examined, with 238 of them belonging to the UK (18.2 Gt in oil and gas fields, 111 Gt in Bunter formations, 26 in Brent formations and 82 Gt in the other formations studied), 3.8 Gt can be stored in Denmark's oil and gas fields, with a rate of 127 Mt $yr^{-1}$, 48 Gt can be stored in Norway's saline aquifers, 1.64 Gt in Germany's 3 potential sites identified and a total of 147 Gt in the Netherlands oil and gas fields, as can be seen in Table 5, or visually in Fig. 15.

Considering the average $CO_2$ emissions of those countries, in the aquifers studied, the UK and the Netherlands can cover 20 times their current carbon footprint, Denmark more than twice, while Norway more than 12 times. Last but not least, the storage efficiency was estimated in the selected fields providing an average of 3.9% in UK, 2% and 2.7% in Denmark and Norway respectively, 3.4% in Germany and 4% in the Netherlands.

These numbers are subject to several assumptions both on the storage model and the available data. The latter are difficult to acquire as they are a product of down-hole measurements of the rocks and are accompanied with significant uncertainties. As for the storage model, the storage capacity estimation is calculated on a pressure basis, meaning the pressure build-up from every injection well is calculated based on the $CO_2$ flowrate. In this way, storage capacity is time-dependent.

These numbers provide us with a clear outcome, considering even the potential uncertainties and errors, which is that extensive application of carbon capture and storage, can be effective in putting $CO_2$ underground. In this way, not only can we limit the increase in the atmospheric $CO_2$ concentration, but also provide ourselves with essential time to implement other, more time demanding, mitigation strategies in order to avoid global warming greater than 1.5 °C by the end of the century.

## CRediT authorship contribution statement

**Panagiotis Karvounis:** Software, Writing – original draft, Investigation, Methodology, Data curation. **Martin J. Blunt:** Conceptualization, Writing – review & editing, Supervision, Methodology, Data curation.

## Declaration of Competing Interest

The authors declare no conflict of interest.

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